

Numerical Experiment Setup for Differentiated Pricing

1 Numerical Experiment Setup

The parameters listed in the numerical setup are sufficient to reproduce the paper-style experiment, as long as the transmission rates and the balanced offloading amount m_k are explicitly computed.

The available system and user parameters are:

$$B, N_0, h_k, F_k, C_k, R_k, p_k, P_{B,k}, \bar{F}, \alpha_k, f_C.$$

From these parameters, the following quantities can be computed:

1. uplink transmission rate r_k^u ,
2. downlink transmission rate r_k^d ,
3. balanced offloading amount m_k ,
4. knapsack weight $w_k = m_k C_k$,
5. knapsack value $v_k = \frac{m_k C_k}{F_k}$,
6. optimal selected users x_k^* ,
7. optimal differentiated price $\mu_k^* = \frac{1}{F_k}$.

1.1 Additional Required Setting

The only additional setting that must be specified is the number of users:

$$K.$$

For a small global-solver test, one can use

$$K = 5$$

or

$$K = 10.$$

To follow the paper's larger simulation scenario, one can use

$$K = 30.$$

The bandwidth is assumed to be equally allocated among the users. Therefore, the bandwidth per user is

$$B_k = \frac{B}{K}.$$

This assumption is consistent with the idea that when the number of users increases, each user receives a smaller portion of the available spectrum.

1.2 Unit Conversion

The total bandwidth is

$$B = 1 \text{ MHz} = 10^6 \text{ Hz}.$$

The local CPU frequency of user k is selected from

$$F_k \in \{0.1, 0.2, \dots, 1\} \text{ GHz}.$$

Therefore, in SI units,

$$F_k = F_k \times 10^9 \text{ cycles/s}.$$

The edge cloud CPU frequency is

$$f_C = 100 \text{ GHz} = 100 \times 10^9 \text{ cycles/s}.$$

The input data size is generated as

$$R_k \in [100, 500] \text{ KB}.$$

This is converted to bits as

$$R_k = R_k \times 1024 \times 8.$$

The noise power spectral density is

$$N_0 = -174 \text{ dBm/Hz}.$$

It is converted to W/Hz by

$$N_0 = 10^{\frac{-174-30}{10}}.$$

The channel gain h_k is generated in dB scale and converted into linear scale by

$$h_k = 10^{\frac{h_k^{dB}}{10}}.$$

1.3 Transmission Rate Calculation

The uplink transmission rate of user k is computed as

$$r_k^u = B_k \log_2 \left(1 + \frac{p_k h_k}{N_0 B_k} \right).$$

The downlink transmission rate of user k is computed as

$$r_k^d = B_k \log_2 \left(1 + \frac{P_{B,k} h_k}{N_0 B_k} \right).$$

Here, p_k is the uplink transmit power of user k , while $P_{B,k}$ is the downlink transmit power from the base station or edge cloud to user k .

1.4 Balanced Offloading Amount

The local computing latency is

$$t_k^{loc} = \frac{(R_k - \ell_k)C_k}{F_k}.$$

The offloading-side latency is

$$t_k^{off} = \frac{\ell_k}{r_k^u} + \frac{\ell_k C_k}{f_C} + \frac{\alpha_k \ell_k}{r_k^d}.$$

Define

$$\beta_k = \frac{1}{r_k^u} + \frac{C_k}{f_C} + \frac{\alpha_k}{r_k^d}.$$

Then the offloading-side latency can be written as

$$t_k^{off} = \beta_k \ell_k.$$

The balanced offloading amount m_k is obtained from the condition

$$t_k^{loc} = t_k^{off}.$$

Therefore,

$$\frac{(R_k - m_k)C_k}{F_k} = \beta_k m_k.$$

Solving for m_k gives

$$m_k = \frac{C_k R_k}{C_k + F_k \beta_k}.$$

This value represents the amount of data that balances the local-computing time and the offloading-side time.

1.5 Knapsack Formulation for Differentiated Pricing

After computing m_k , the differentiated pricing problem can be written as a binary knapsack problem.

The cloud capacity consumed by user k is

$$w_k = m_k C_k.$$

The revenue value obtained from selecting user k is

$$v_k = \frac{m_k C_k}{F_k}.$$

The binary decision variable is

$$x_k \in \{0, 1\},$$

where

$$x_k = \begin{cases} 1, & \text{if user } k \text{ is selected for offloading,} \\ 0, & \text{otherwise.} \end{cases}$$

Therefore, the differentiated pricing optimization problem becomes

$$\max_{x_k \in \{0,1\}} \sum_{k=1}^K v_k x_k$$

subject to

$$\sum_{k=1}^K w_k x_k \leq \bar{F},$$

$$x_k \in \{0, 1\}, \quad \forall k.$$

Substituting w_k and v_k , the problem can also be written as

$$\max_{x_k \in \{0,1\}} \sum_{k=1}^K \frac{m_k C_k}{F_k} x_k$$

subject to

$$\sum_{k=1}^K m_k C_k x_k \leq \bar{F}.$$

1.6 Price Recovery

After the optimal binary decision x_k^* is obtained, the optimal differentiated price is recovered as

$$\mu_k^* = \begin{cases} \frac{1}{F_k}, & x_k^* = 1, \\ \text{sufficiently large price,} & x_k^* = 0. \end{cases}$$

If user k is selected, the cloud charges the maximum price that the user is still willing to accept, namely

$$\mu_k^* = \frac{1}{F_k}.$$

If user k is not selected, the cloud sets a sufficiently high price so that the user chooses not to offload.

1.7 Latency Measurement

After solving the offloading decision, the optimal offloaded data size of user k is

$$\ell_k^* = m_k x_k^*.$$

The local-computation latency becomes

$$t_k^{loc} = \frac{(R_k - \ell_k^*)C_k}{F_k}.$$

The offloading-side latency becomes

$$t_k^{off} = \frac{\ell_k^*}{r_k^u} + \frac{\ell_k^* C_k}{f_C} + \frac{\alpha_k \ell_k^*}{r_k^d}.$$

The total latency of user k is

$$t_k = \max(t_k^{loc}, t_k^{off}).$$

The average latency over all users is

$$\bar{t} = \frac{1}{K} \sum_{k=1}^K t_k.$$

This average latency includes both selected and unselected users. If $x_k^* = 0$, then

$$\ell_k^* = 0,$$

so user k computes all of its task locally and its latency becomes

$$t_k = \frac{R_k C_k}{F_k}.$$

1.8 Effect of Increasing the Number of Users

If the total bandwidth B and the cloud computation capacity $\bar{F}$ remain fixed, increasing the number of users generally increases the average latency.

This is because the bandwidth per user is

$$B_k = \frac{B}{K}.$$

As K increases, B_k decreases. This reduces the uplink and downlink transmission rates r_k^u and r_k^d . Consequently, the transmission components of the offloading latency increase.

Furthermore, more users compete for the same limited cloud computation capacity $\bar{F}$. Therefore, not all users can be selected for offloading. Users that are not selected must compute locally, and if they have weak local CPU capability, their local latency can be high.

Therefore, with fixed system resources, the expected trend is

$$K \uparrow \Rightarrow B_k \downarrow \Rightarrow r_k^u, r_k^d \downarrow \Rightarrow \bar{t} \uparrow.$$

However, because user parameters are randomly generated, the increase in average latency may not be strictly monotonic in every single simulation run. For example, if newly added users have small task sizes R_k , low computation intensity C_k , or strong local CPU frequency F_k , the average latency may not always increase for that particular realization.

1.9 Summary

The parameters in the numerical setup are sufficient to reproduce the experiment because they allow the generation of each user's computation capability F_k , task size R_k , task intensity C_k , and wireless channel condition h_k . These parameters are used to compute the uplink and downlink transmission rates, the balanced offloading amount m_k , the knapsack weight $w_k = m_k C_k$, and the knapsack value $v_k = m_k C_k / F_k$.

The only additional setting required is the number of users K . With the assumption of equal bandwidth allocation $B_k = B/K$, the differentiated-pricing problem can then be solved using a global solver such as SCIP, BARON, or Couenne.